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Cross-modal Content-based Retrieval for 3D Medical Images

Neurosurgical AI as a motivating example

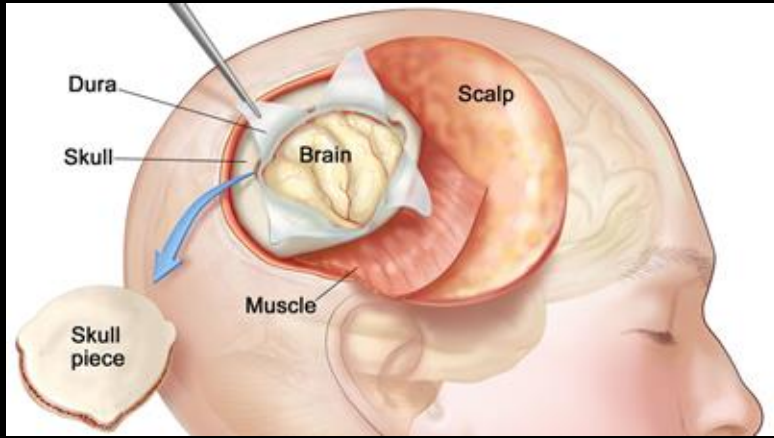
Clinical objectives

Surgical resection is the critical first step for treating most brain tumors.

Competing surgical objectives:
- **Maximal tumor removal** to increase the overall survival.
- **Minimal functional damage** to protect the patient's quality of life.

Research aim:
Develop **AI methods** to help neurosurgeons improve surgical outcomes.

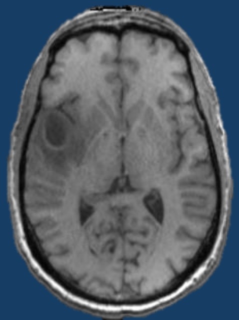
Craniotomy



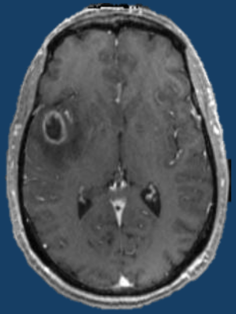
Medical data is multimodal

Tumorous areas:

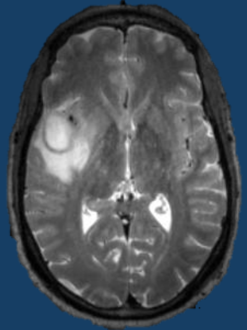
Structural MRI



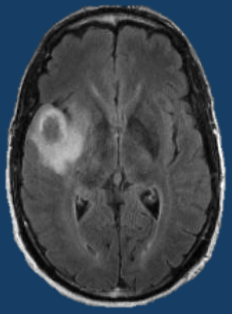
T1



Contrast-enhanced T1



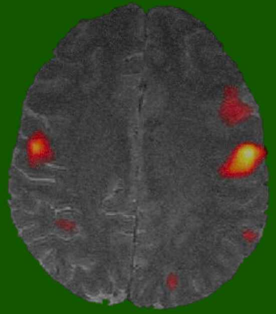
T2



FLAIR

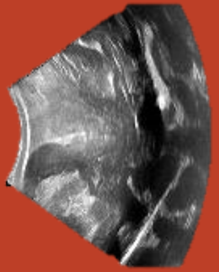
Functional areas:

Functional MRI



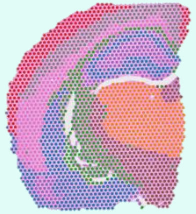
Intraoperative deformations:

Ultrasound

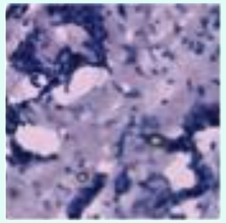


Biological characteristics:

Omics



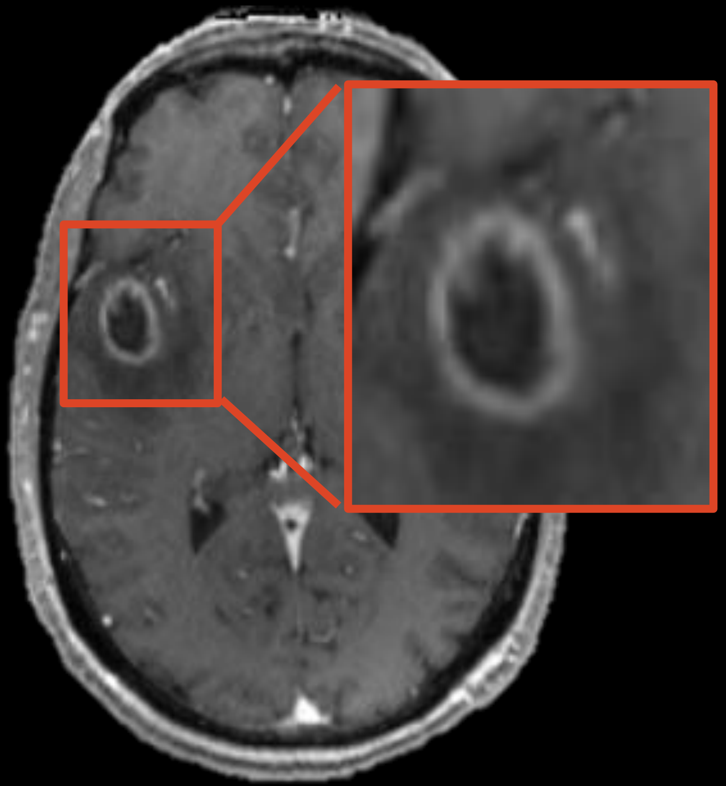
Histopathology



Need to **combine multimodal information** to enable a personalized surgical procedure.

Medical Image Interpretation

Perception



Analysis

Glioblastoma - WHO Grade IV
→ Aggressive Tumor
→ 14–16 months survival

Right frontoparietal region
→ Primary motor cortex
→ Primary Somatosensory cortex (pain, temperature,)

Recommendation

Step 1: Surgical resection
→ Maximal safe resection
→ If possible: awake craniotomy with cortical mapping

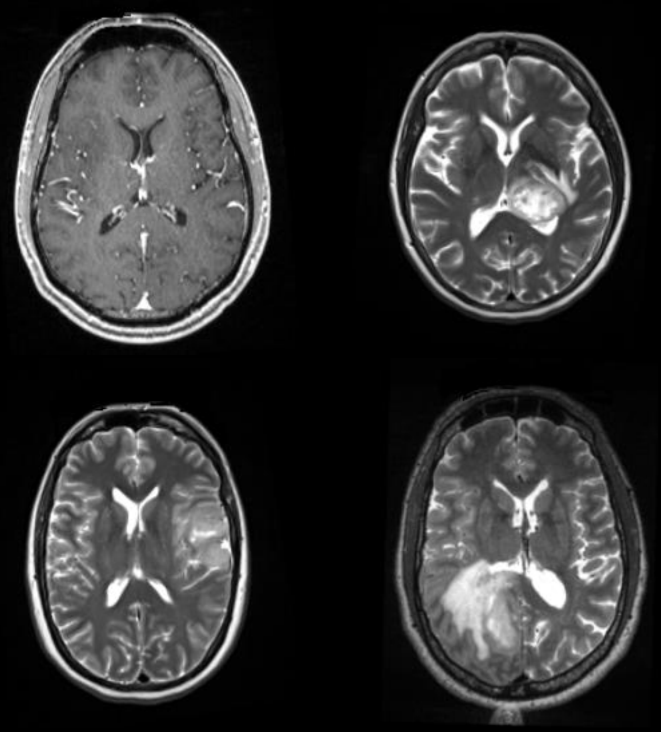
Step 2: Radiotherapy after surgery

Medical image interpretation is based on experience.

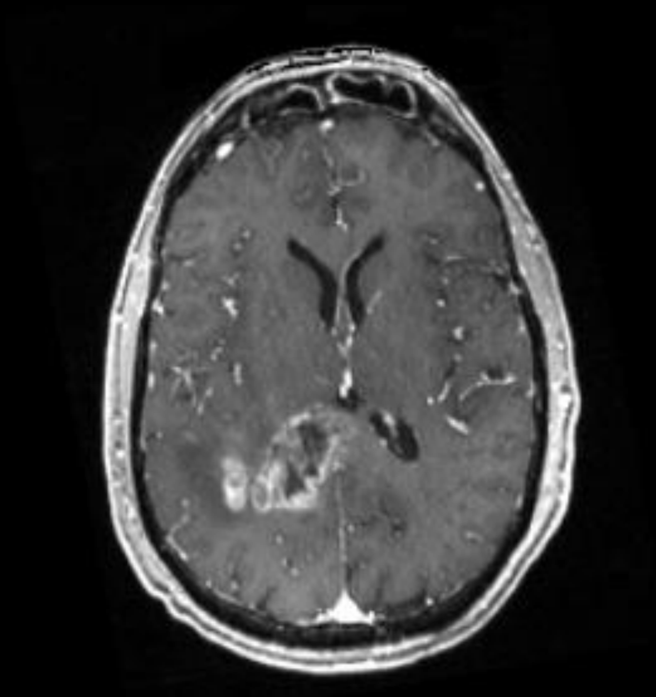
Content-based retrieval

Need to **automatically** identify similar cases based on image context.

Previous cases



New case



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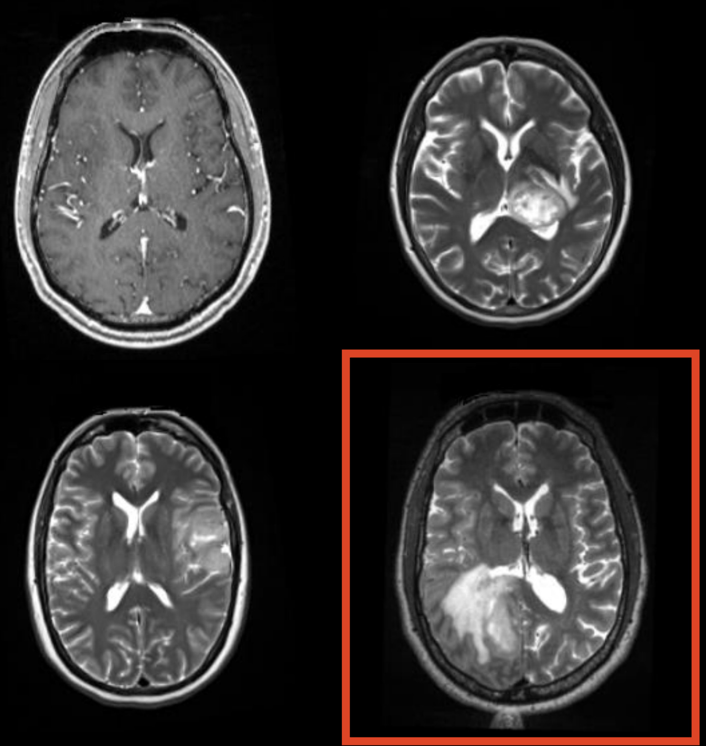
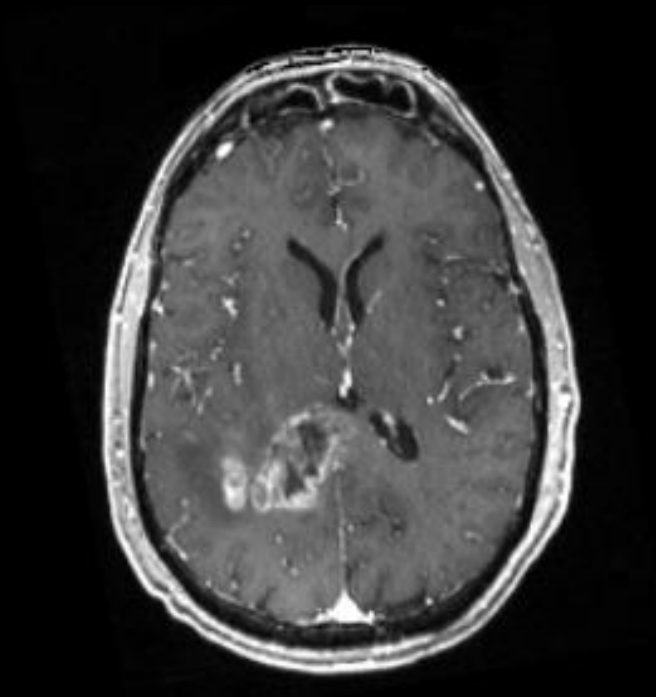


New case



Survival: 15 months
Cognitive deficits after surgery

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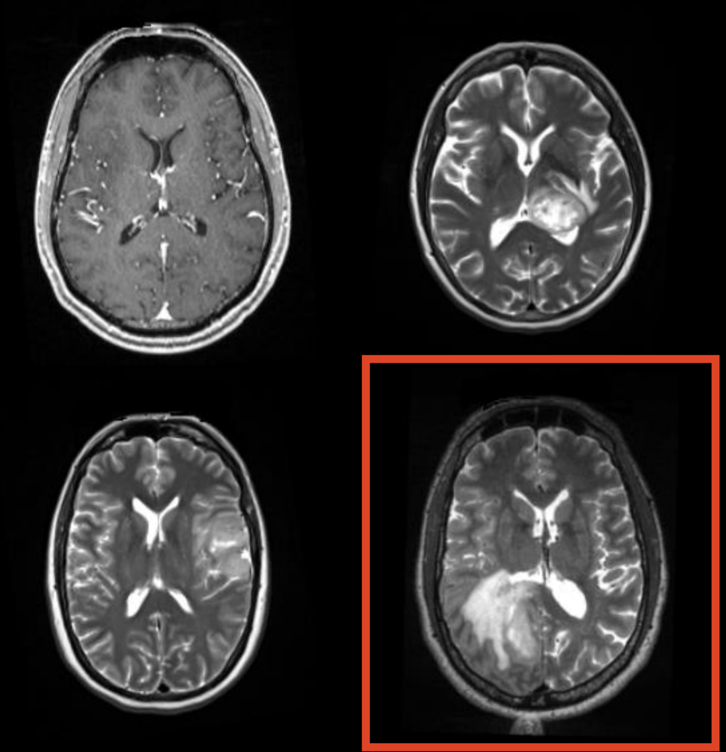
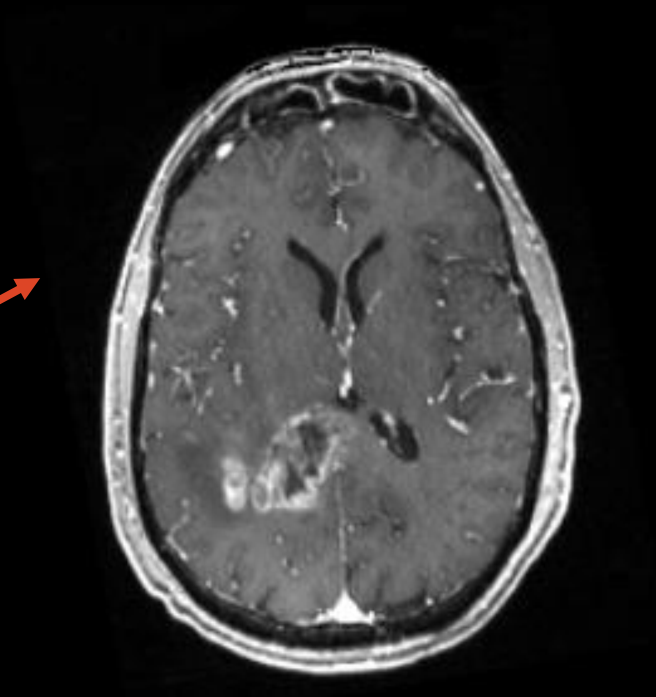


New case



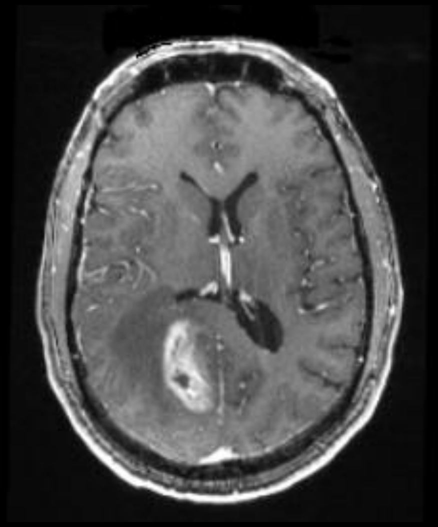
Modality gap

Survival: 15 months
Cognitive deficits after surgery

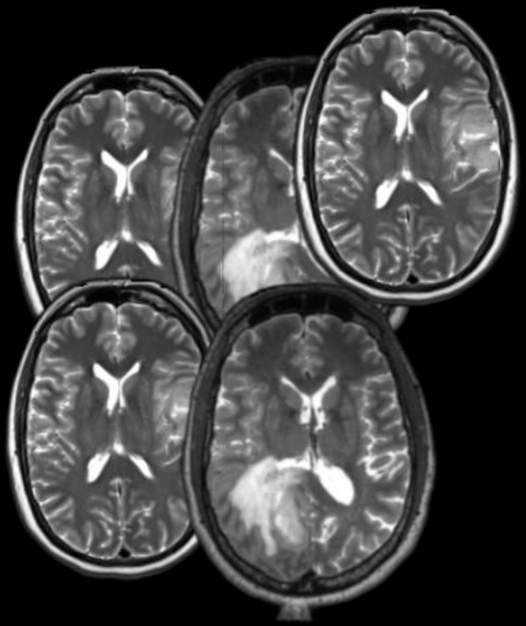
Objective of the challenge

Given a 3D MRI (contrast-enhanced T1) and a set of target images (T2)
Find the T2 image from the same patient.

Query: Contrast-enhanced T1



Gallery: T2 images

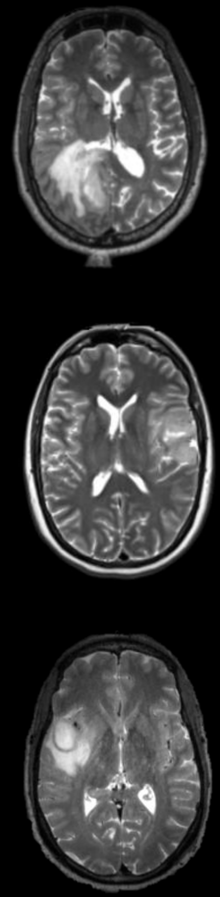


Predict

Rank 1

Rank 2

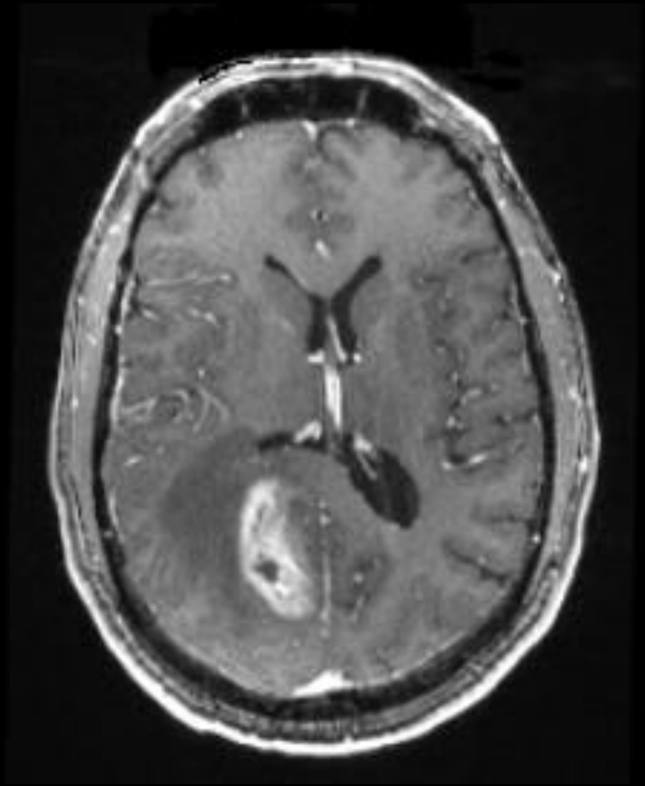
Rank 3



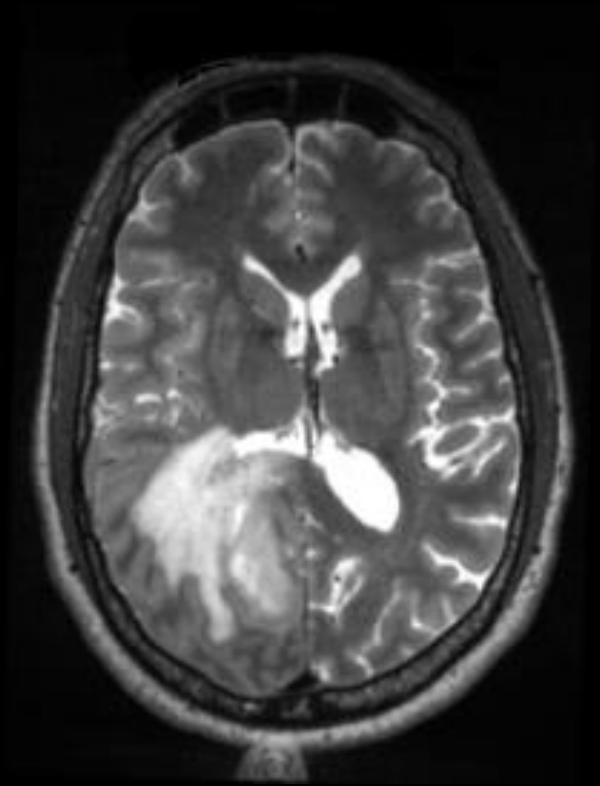
...

Level 1: perfectly aligned

Input: ceT1

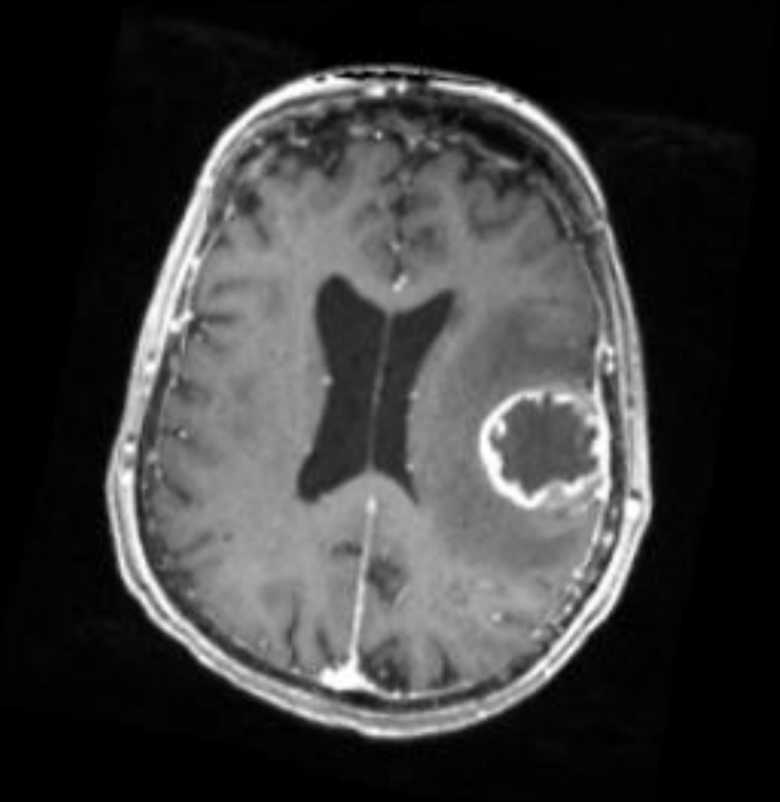


Target: T2

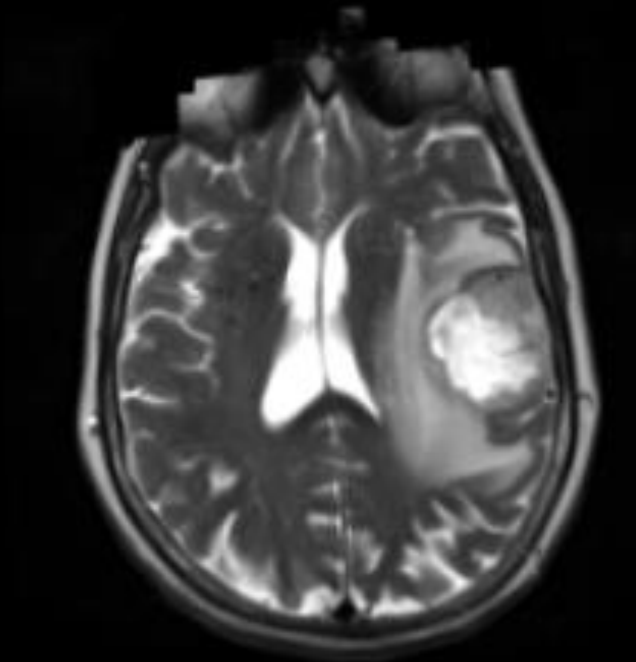


Level 2: non-linear deformations

Input: ceT1

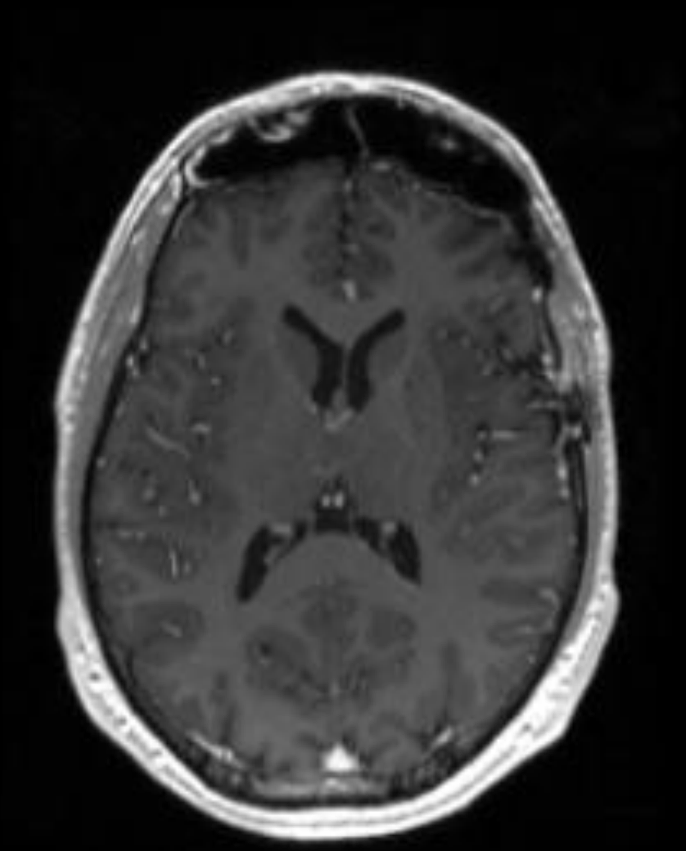


Target: T2

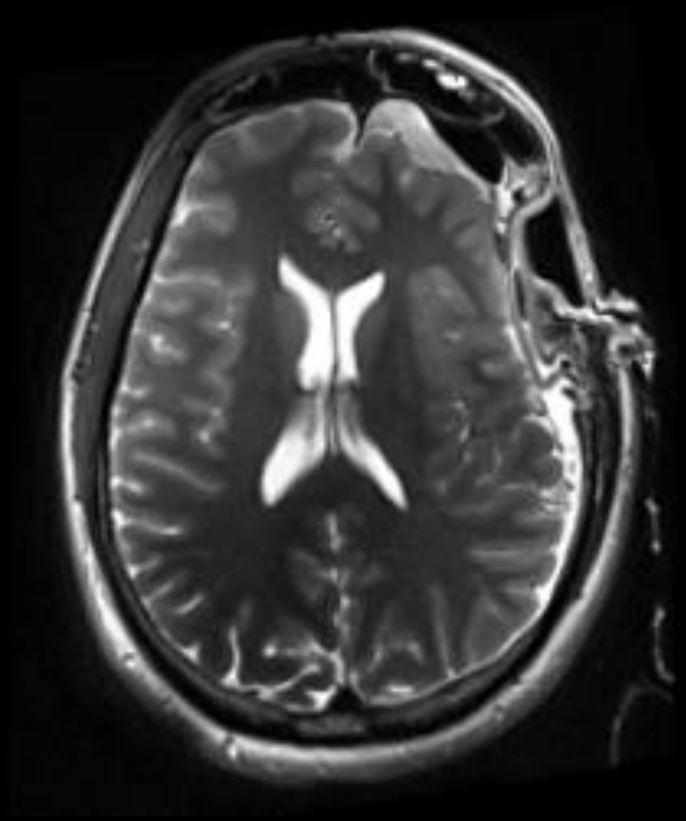


Level 3: before / after surgery

Input: ceT1



Target: T2



+ test images from a different hospital!

Dataset and evaluation

Training:

- 350 paired images (ceT1 + T2)

Validation/Test

- 3 datasets (one per level)
- For each query image, rank all the target images from the gallery from the most similar to the least similar
- Evaluation per dataset:

  best rank = score 1; worst rank = score 0 (higher is better)
- Final metric: Macro-average (weighted average per dataset)

Get started!

Public repo:
https://github.com/NicoStellwag/ehl-paris-2026-medical-retrieval

Contains:
- Data description
- Evaluation design
- Baseline code
- Submission on Kaggle

Thanks!